Soy Milk Plant



Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Soy milk plant

What it is used for

A soy milk plant processes soybeans into soy milk through cleaning, soaking, grinding, heating and filtration. Proper operation supports consistent product quality, food hygiene and safe handling of hot liquids and machinery.

Main parts

- Identify the soybean receiving hopper and cleaning section.
- Locate the soaking tank and water supply arrangement.
- Identify the grinder or wet milling unit.
- Locate the heating vessel or cooking system.
- Identify the filtration unit, filter cloth or separation mechanism.
- Locate the transfer pumps, pipes and product collection tank.
- Identify temperature controls, emergency stops and cleaning access points.

Before you start

1. Read the operating instructions and follow food hygiene procedures.
2. Inspect tanks, grinders, pipes, valves, guards and electrical connections for damage.
3. Clean and sanitise all food-contact surfaces and utensils.
4. Select suitable soybeans and remove stones, damaged beans and foreign materials.
5. Soak beans for the approved time using clean water and hygienic containers.
6. Check that the heating, filtration and transfer systems are correctly assembled.
7. Wear clean protective clothing and heat-resistant gloves where required.

How to use

1. Clean and sort the soybeans using the designated cleaning section.
2. Soak the beans according to the approved processing instructions.
3. Drain soaking water and rinse beans where the process requires it.
4. Feed prepared beans and the specified quantity of water into the grinder.
5. Operate the grinder with all guards secured and collect the resulting slurry.
6. Transfer the slurry to the designated heating vessel.
7. Heat the mixture according to the validated process and stir as required.
8. Filter the processed mixture using the specified filtration arrangement.
9. Transfer the soy milk into a clean, sanitised collection container.
10. Cool or further process the product promptly according to food safety instructions.
11. Stop the equipment and isolate power before dismantling or cleaning components.

⚠ Safety

- Keep hands away from grinders, agitators and other moving parts.
- Avoid contact with hot soy milk, steam and heated vessel surfaces.
- Never open hot pressurised equipment unless the approved procedure permits it.
- Disconnect power before clearing blockages or cleaning internal machinery.
- Prevent cross-contamination by separating raw beans, waste and finished soy milk.
- Follow validated heat treatment and hygienic storage requirements; do not assume filtration alone makes the product safe.

Care & storage

- Remove soybean residue promptly from grinders, filters, pipes and tanks.
- Wash, rinse and sanitise all food-contact components after processing.
- Inspect blades, seals, hoses and electrical connections for wear or damage.
- Dry removable parts and store them in a clean, protected area.

Common mistakes to avoid

- Using spoiled beans or contaminated soaking water.
- Overloading the grinder or using an unsuitable water ratio.
- Failing to apply the required heat treatment or hygienic cooling.
- Leaving soy residue in pipes, filters or tanks after production.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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